Comprehensive Documentation: Pansharpening Image Fusion using Multi-Scale Wavelet Vision Transformers (HWViT) and NSGA-II

Project Documentation

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Abstract

This document outlines the entire architecture, pipeline, mathematical formulations, and optimization strategy for the Multispectral and Panchromatic Image Fusion (Pansharpening) project. The proposed model, named HWViT, integrates Discrete Wavelet Transforms (DWT) with attention mechanisms to effectively fuse high-spatial-resolution Panchromatic (PAN) images and high-spectral-resolution Multispectral (MS) images. Furthermore, the Non-dominated Sorting Genetic Algorithm II (NSGA-II) is employed to optimize hyperparameters for a multi-objective Pareto front.

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1 Introduction

Pansharpening is an image fusion technique that combines a high-resolution, single-band Panchromatic (PAN) image with a lower-resolution, multi-band Multispectral (MS) image. The goal is to generate an image that possesses both the high spatial resolution of the PAN image and the high spectral resolution of the MS image. This project implements a novel deep learning architecture based on Wavelet Vision Transformers and optimizes the architectural hyperparameters using a multi-objective genetic algorithm (NSGA-II).

2 Proposed Architecture: HWViT

The core architecture is the Hierarchical Wavelet Vision Transformer (HWViT). It operates on PAN, MS, and upsampled MS (LMS) images to fuse information progressively at both fine and large scales.

2.1 Overall Pipeline

The overall pipeline comprises the following steps:

1. **Feature Extraction:** The PAN image undergoes channel projection using a continuous convolutional block (`raise_channel`) to match `pan_target_channel`. The MS image is feature-extracted and upsampled via a PixelShuffle layer.

2. **Scale processing:**

   - **L-MWiT Block (Large Scale):** Operates on downsampled PAN and LMS features to capture low-frequency structural components.
   - **F-MWiT Block (Fine Scale):** Takes the full-resolution projected PAN, partially downsampled LMS, and the output of the L-MWiT block to inject high-frequency details.

3. **Feature Reduction and Combination:** The fused output from the F-MWiT block is projected back to the original LMS channel dimension and added to the initial upsampled MS features using a residual connection.

2.2 Multi-Scale Wavelet Vision Transformer (MWiT)

The `S_MWiT` (Sub-MWiT) module, utilized by both L-MWiT and F-MWiT, forms the foundation of the fusion process. Its operation is defined as follows:

2.2.1 Discrete Wavelet Transform (DWT)

The DWT isolates the image into approximation and detail subbands. Using fixed 2D Haar Wavelet filters, an input x is decomposed into LL (low-frequency), LH (horizontal), HL (vertical), and HH (diagonal) components. The forward Haar DWT filters are defined mathematically as:

$$w_{LL} = \frac{1}{4}\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \quad w_{LH} = \frac{1}{4}\begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}$$
$$w_{HL} = \frac{1}{4}\begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}, \quad w_{HH} = \frac{1}{4}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

Using stride 2 convolutions, the spatial dimensions are halved while retaining all spatial and spectral semantics within the combined subbands. The inverse transform (IDWT) reconstructs the full-size images from these subbands using transposed convolutions.

2.2.2 Cross-Attention Mechanism

Once the PAN input is decomposed into DWT subbands $wd_{LL}, wd_{LH}, wd_{HL}, wd_{HH}$, they are mapped to the MS feature dimension (L_{up}). A combined latent representation v is acquired by averaging the MS input, PAN features, and contextual background image information using learned parameters α, β:

$$v = \text{ResBlock}(\alpha \cdot x_{\text{PAN}} + \beta \cdot x_{\text{MS}} + (1 - \alpha - \beta) \cdot \text{FFN}(x_{\text{background}}))$$

The attention mechanism processes these representations. For each DWT component $i \in \{LL, LH, HL, HH\}$:

$$\begin{aligned} Q_i &= \text{Linear\_Q}(\text{LayerNorm}(wd_{i,\text{MS}})) \\ K_i &= \text{Linear\_K}(\text{LayerNorm}(wd_{LL,\text{MS}})) \\ V &= \text{Linear\_V}(\text{LayerNorm}(v)) \\ \text{Attn}_i &= \text{Softmax}\left(\frac{Q_i K_i^T}{\sqrt{d_k}}\right) V \end{aligned}$$

Here, the LL subband of the PAN image serves as the key (K_i) for all other subbands, ensuring structural consistency across frequency domains. The responses Attn_i are later concatenated and reconstructed via IDWT.

2.3 Feed-Forward Network (FFN) and Resblocks

The architectural flow extensively leverages Resblocks and Feed-Forward Networks (`FFN` and `FFN_2`). The regular `FFN` scales the channel dimensions down to a bottleneck ($C/2$) and back up to project features effectively. It involves:

$$x_{\text{mid}} = \text{PReLU}(\text{Conv}(x_{\text{in}}))$$

$$x_{\text{out}} = \text{Linear}_2(\text{Conv}_{1\times 1}(x_{\text{mid}}) + x_{\text{in}})$$

3 Training Pipeline

3.1 Dataset Configuration

The models are trained using HDF5 datasets containing:

- `pan`: Panchromatic image inputs.
- `ms`: Low-resolution Multispectral inputs.
- `lms`: Upsampled Multispectral baseline.
- `gt`: Ground Truth image arrays.

An 80/20 train-validation split is enforced dynamically on dimensions of 64×64 patches. Mini-batching is employed (batch size 8). The input intensities form tensors in the continuous range following radiometric normalization controlled by 'ratio' (e.g., 2047).

3.2 Loss Function

The training algorithm updates gradients purely based on **L1 Loss (Mean Absolute Error)**:

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^{N} |\hat{y}_i - y_i|$$

The optimizer used is Adam, evaluated iteratively over the specified number of epochs.

4 Non-Dominated Sorting Genetic Algorithm II (NSGA-II)

NSGA-II (Non-Dominated Sorting Genetic Algorithm II) is one of the most widely used multi-objective evolutionary algorithms for solving optimization problems involving two or more conflicting objectives. It was proposed by *Deb et al.* in 2002 as an improvement over the original NSGA algorithm.

The main goal of NSGA-II is to obtain a set of Pareto-optimal solutions that represent the trade-offs among conflicting objectives.

Suppose a multi-objective optimization problem is written as:

$$\min \; F(x) = [f_1(x), f_2(x), \ldots, f_M(x)] \tag{1}$$

subject to:

$$g_j(x) \leq 0, \quad j = 1, 2, \ldots, J \tag{2}$$

$$h_k(x) = 0, \quad k = 1, 2, \ldots, K \tag{3}$$

where:

- $x = [x_1, x_2, \ldots, x_n]$ is the decision vector,
- $f_i(x)$ are the objective functions,
- M is the number of objectives,
- $g_j(x)$ are inequality constraints,
- $h_k(x)$ are equality constraints.

4.1 Pareto Dominance

NSGA-II is based on the concept of Pareto dominance.

A solution $x^{(1)}$ is said to dominate another solution $x^{(2)}$ if:

1. $x^{(1)}$ is no worse than $x^{(2)}$ in all objectives:

$$f_i(x^{(1)}) \leq f_i(x^{(2)}), \quad \forall i \tag{4}$$

2. $x^{(1)}$ is strictly better than $x^{(2)}$ in at least one objective:

$$\exists i \text{ such that } f_i(x^{(1)}) < f_i(x^{(2)}) \tag{5}$$

The set of solutions that are not dominated by any other solution is called the Pareto-optimal set.

4.2 Main Steps of NSGA-II

NSGA-II works iteratively through generations. At each generation, it performs the following major steps:

1. Initialize a random population.
2. Evaluate objective functions.
3. Perform non-dominated sorting.
4. Compute crowding distance.
5. Use tournament selection.
6. Apply crossover and mutation.
7. Combine parent and offspring populations.
8. Select the next generation.

4.3 Population Initialization

Assume the population size is N. The initial population is:

$$P_0 = \{x_1, x_2, \ldots, x_N\} \tag{6}$$

Each solution is evaluated for all objective functions.

4.4 Fast Non-Dominated Sorting

The most important feature of NSGA-II is fast non-dominated sorting.

For every solution p, define:

- S_p: the set of solutions dominated by p
- n_p: the number of solutions that dominate p

Initially:

$$n_p = 0 \tag{7}$$

For each pair of solutions (p, q):

- If p dominates q, then add q to S_p
- If q dominates p, then increment n_p

After checking all pairs:

- All solutions with $n_p = 0$ belong to the first front F_1
- Remove F_1 temporarily and repeat the process to obtain $F_2, F_3, \ldots$

Thus:

$$F_1, F_2, F_3, \ldots \tag{8}$$

represent successive Pareto fronts.
Solutions in lower front numbers are considered better.

4.5 Crowding Distance

Since many solutions may lie in the same Pareto front, NSGA-II uses crowding distance to maintain diversity.

For each objective m, sort the solutions in a front according to objective value f_m.
The crowding distance of the i-th solution is calculated as:

$$CD_i = \sum_{m=1}^{M} \frac{f_m^{i+1} - f_m^{i-1}}{f_m^{\max} - f_m^{\min}} \tag{9}$$

where:

- f_m^{i+1} is the next objective value,
- f_m^{i-1} is the previous objective value,
- $f_m^{\max}$ and $f_m^{\min}$ are the maximum and minimum values of objective m.

Boundary solutions are assigned infinite crowding distance:

$$CD = \infty \tag{10}$$

A larger crowding distance means the solution lies in a less crowded region of the Pareto front.

4.6 Crowded Tournament Selection

NSGA-II uses binary tournament selection.

Given two candidate solutions, preference is given according to:

1. Lower rank (better Pareto front)
2. Higher crowding distance if ranks are equal

Thus, for two solutions p and q:

$$p \prec q \tag{11}$$

if either:

$$\text{rank}(p) < \text{rank}(q) \tag{12}$$

or:

$$\text{rank}(p) = \text{rank}(q) \quad \text{and} \quad CD_p > CD_q \tag{13}$$

4.7 Crossover and Mutation

After selection, offspring are generated using crossover and mutation.

A common crossover operator in NSGA-II is Simulated Binary Crossover (SBX).

For parent solutions x_1 and x_2, offspring are generated as:

$$\beta_q = \begin{cases} (2u)^{\frac{1}{\eta_c+1}}, & u \le 0.5 \\ \left(\frac{1}{2(1-u)}\right)^{\frac{1}{\eta_c+1}}, & u > 0.5 \end{cases} \tag{14}$$

where:

- $u \in [0, 1]$ is a random number,
- η_c is the crossover distribution index.

The offspring are:

$$child_1 = 0.5[(1 + \beta_q)x_1 + (1 - \beta_q)x_2] \tag{15}$$

$$child_2 = 0.5[(1 - \beta_q)x_1 + (1 + \beta_q)x_2] \tag{16}$$

Polynomial mutation is also commonly used:

$$\delta_q = \begin{cases} (2r)^{\frac{1}{\eta_m+1}} - 1, & r < 0.5 \\ 1 - [2(1-r)]^{\frac{1}{\eta_m+1}}, & r \ge 0.5 \end{cases} \tag{17}$$

where:

- $r \in [0, 1]$ is a random number,
- η_m is the mutation distribution index.

The mutated variable becomes:

$$x' = x + \delta_q(x_{\max} - x_{\min}) \tag{18}$$

4.8 Elitism in NSGA-II

NSGA-II uses elitism to preserve good solutions.

Suppose:

$$|P_t| = N \tag{19}$$

and offspring population is:

$$|Q_t| = N \tag{20}$$

They are combined into:

$$R_t = P_t \cup Q_t \tag{21}$$

Thus:

$$|R_t| = 2N \tag{22}$$

The combined population is sorted into fronts:

$$F_1, F_2, F_3, \ldots \tag{23}$$

The next generation population P_{t+1} is filled front by front until size N is reached.

If the last front cannot be completely included, solutions with the highest crowding distance are selected.

4.9 Computational Complexity

The computational complexity of NSGA-II is mainly due to non-dominated sorting.

For a population of size N and M objectives, the complexity is:

$$O(MN^2) \tag{24}$$

This is significantly better than the original NSGA algorithm.

4.10 Advantages of NSGA-II

- Efficient non-dominated sorting
- Maintains diversity using crowding distance
- Preserves elite solutions
- Produces a well-distributed Pareto front
- Widely used for engineering, image fusion, machine learning, and resource allocation problems

4.11 NSGA-II for Image Fusion Parameter Optimization

In image fusion or pansharpening applications, NSGA-II can optimize parameters of a fusion method by simultaneously maximizing spectral quality and spatial quality.

For example, objectives may include:

$$\max\ QNR \tag{25}$$

$$\min\ SAM \tag{26}$$

$$\min\ ERGAS \tag{27}$$

$$\max\ SCC \tag{28}$$

where:

- QNR = Quality with No Reference,
- SAM = Spectral Angle Mapper,
- ERGAS = Relative Global Dimensional Error,
- SCC = Spatial Correlation Coefficient.

Thus, NSGA-II finds parameter sets that provide the best trade-off between preserving spectral information and enhancing spatial details.

5 NSGA-II Multi-Objective Optimization

Deep learning architectures for pansharpening trade off varying perceptual metrics at the cost of computational complexity. To automate this design search, the Non-dominated Sorting Genetic Algorithm II (NSGA-II) isolates the optimal Pareto Front.

5.1 Hyperparameter Search Space

The variables defining an individual in the Genetic Algorithm population are:

1. `pan_raw`: Approximate channel dimension for PAN tracking $\in [16, 64]$.
2. `ms_raw`: Approximate channel dimension for MS tracking $\in [16, 64]$.
3. `head_channel`: Dimensions utilized per Attention head $\in [4, 16]$.
4. `dropout`: Probability of dropout in network $\in [0.01, 0.3]$.
5. `lr`: Adam learning rate $\in [10^{-4}, 10^{-3}]$.

Note: `pan_target_channel` and `ms_target_channel` are rounded to the nearest integer multiple of `head_channel` to ensure equal dimension slicing during attention computation.

5.2 Target Objectives (Fitness Constraints)

NSGA-II minimizes an array of 7 mathematical objectives. Signs are flipped internally in code where a metric traditionally requires maximization.

5.2.1 1. Spectral Angle Mapper (SAM) - Minimize

Computes the spectral deviation angle between the ground truth (y) and predicted ($\hat{y}$) image.

$$\text{SAM} = \arccos\left(\frac{\sum \hat{y} \cdot y}{\|\hat{y}\|_2 \|y\|_2 + \epsilon}\right)$$

5.2.2 2. Erreur Relative Globale Adimensionnelle de Synthèse (ERGAS) - Minimize

Tracks quantitative global error per spectral channel scaled by spatial resolution.

$$\text{ERGAS} = \frac{100}{r}\sqrt{\frac{1}{C}\sum_{i=1}^{C}\left(\frac{\text{RMSE}_i}{\overline{y}_i}\right)^2}$$

where $r = 4$ is the resolution ratio, and $\overline{y}_i$ is the mean of the i-th band.

5.2.3 3. Correlation Coefficient (CC) - Maximize

Indicates the linear spectral phase dependency.

$$\text{CC} = \frac{\text{Cov}(\hat{y}, y)}{\sigma_{\hat{y}}\sigma_y}$$

5.2.4 4. Standard Deviation Difference (SD) - Minimize

Evaluates variation disparity representing noise distribution differences.

$$\text{SD} = |\sigma_{\hat{y}} - \sigma_y|$$

5.2.5 5. Spatial Frequency (SF) - Maximize

Quantifies the overall active spatial frequency map reflecting image sharpness.

$$\text{SF} = \sqrt{(\text{Row Frequency})^2 + (\text{Column Frequency})^2}$$

where frequencies correlate to root-mean-square differences between adjacent pixels over rows and cols respectively.

5.2.6 6. Structural Similarity Index (SSIM) - Maximize

Verifies perceptual structural similarities over sliding windows.

$$\text{SSIM} = \frac{(2\mu_{\hat{y}}\mu_y + c_1)(2\sigma_{\hat{y}y} + c_2)}{(\mu_{\hat{y}}^2 + \mu_y^2 + c_1)(\sigma_{\hat{y}}^2 + \sigma_y^2 + c_2)}$$

5.2.7 7. Network Parameters (`n_params`) - Minimize

Sums total trainable parameter count in the HWViT instantiation. Ensures efficiency for downstream inference without exorbitant deployment cost.

6 Results and Performance Analysis

Following the multi-objective optimization phase using the NSGA-II algorithm, the Pareto front was extracted to understand the trade-offs between various structural, spectral, and computational objectives. The solutions generated by the model are visualized across multiple metric dimensions. Detailed analysis of these visualizations yields critical insights into the performance behavior of the proposed HWViT architecture.

6.1 Overall Pareto Front Distribution

The **Parallel Coordinates Plot** maps the objective values (SAM, ERGAS, CC, SD, SF, SSIM, and total parameters) for all candidate models across the Pareto front.

- **Scale Discrepancy:** It is immediately apparent that the number of trainable parameters (*n_params*) dominates the scale of the plot, spanning from near zero to approximately 1.4×1.6^6 (1.4 million) parameters. In contrast, image quality metrics (such as SAM and ERGAS) fall within a much smaller numerical range (e.g., SAM $\in [0.09, 0.18]$), making them appear as a flat horizontal line near zero on a raw scale graph.
- **Computational Trade-off:** The wide distribution of parameter counts indicates that the primary conflict driving the Pareto front diversity is the trade-off between model complexity (computational cost) and image quality. The NSGA-II solver successfully explored a vast architectural space, finding highly lightweight models alongside much deeper, heavier networks.

6.2 Spectral Metrics: SAM vs. ERGAS

Analysis of the 2D scatter plot mapping **SAM against ERGAS** reveals a strong positive correlation between these two objectives.

- **Synergistic Objectives:** As the Spectral Angle Mapper (SAM) distortion increases from approximately 0.09 to 0.18, the ERGAS error proportionally worsens, scaling from roughly 4.0 to 8.1.

- **Inference:** Since both metrics measure aspects of spectral and global spatial-spectral fidelity, they are not strictly conflicting objectives. A model that preserves the spectral angle well also tends to minimize the global radiometric error. The spread of solutions along this positive linear trajectory confirms that moving along the Pareto front primarily shifts the overall quality/complexity trade-off rather than trading ERGAS for SAM.

6.3 Spatial-Spectral Trade-off: SAM vs. Spatial Frequency (SF)

The relationship between **SAM and Spatial Frequency (SF)** demonstrates greater scattering and a more complex manifold.

- **Metric Tension:** While better (lower) SAM values are clustered on the left side of the axis, the corresponding Spatial Frequency (which should be maximized for perceptually sharper images) varies widely between 0.0825 and 0.100.
- **Inference:** This plot captures the classic pansharpening spatial-spectral trade-off. Certain architectures on the front achieve very high spatial frequencies (extreme sharpness) but suffer from noticeable spectral distortion (higher SAM). Conversely, architectures that maintain near-perfect spectral fidelity sometimes produce smoother images with lower high-frequency spatial transitions.

6.4 3D Performance Manifold: SAM, ERGAS, and SSIM

The **3D Pareto Front** visualization maps SAM, ERGAS, and the Structural Similarity Index Measure (SSIM) simultaneously.

- **Linear Quality Manifold:** The solutions do not form a scattered cloud in this 3D sub-space; instead, they trace a distinct, curved uni-dimensional trajectory.
- **Inference:** This demonstrates a high degree of agreement among the perceptual image quality metrics. The optimal models seamlessly reach a peak state where SAM is minimized (~ 0.09), ERGAS is minimized (~ 4.0), and SSIM is maximized simultaneously (~ 0.975). Conversely, models generated with high error margins score poorly across all three metrics concurrently.
- **Conclusion on Optimization:** Because these standard image quality metrics align so uniformly on the discovered front, it proves that the true "multi-objective" conflict solved by NSGA-II is entirely between *spatial sharpness/spectral fidelity* and *model size/efficiency*. Researchers can simply slide along this continuous curve to choose exactly how much parameter overhead they are willing to accept for a proportional, guaranteed increase in overall visual quality.

7 Conclusion

The integration of a hierarchical multi-scale DWT approach alongside self-attention offers precise pansharpening capabilities while minimizing spectral distortion. By wrapping the network layout in the NSGA-II solver, users can derive lightweight models, highly structurally-accurate networks, or heavily balanced optimal designs by visualizing the resultant 7D Pareto Front.